

Personal Health OS

Complete Product Documentation for Vendor Quotation

Business Model + BRD + PRD + Feature Breakdown + Estimation Guidance

Document Type	Master Product Documentation / Quotation Version
Prepared For	Software Houses, Product Teams, CTOs, Estimation Teams
Primary Language	English
Primary Platforms	Mobile Apps (iOS / Android) + Web Doctor Viewer
Business Model	Hybrid Subscription + Credits
Version	Final Reviewed Draft

This document is intended to give development vendors enough clarity to provide a structured technical and commercial quotation.

1. Executive Summary

Personal Health OS is a mobile-first digital health platform designed to help individuals and families organize, track, and present their medical history through a conversational experience supported by AI. The product combines document storage, structured health tracking, reminders, symptom capture, timeline visualization, and doctor-ready reporting in one system.

The product is intentionally designed as a health record organization and communication platform, not as a diagnostic or treatment decision system. Its most important differentiator is the ability for a user to generate an AI-powered doctor report for any chosen date range and present it through a doctor-friendly web view or QR access during a consultation.

Quick Summary for Readers Who Do Not Want the Full Detail

Area	Short Summary
Platforms	Mobile apps for iOS and Android, plus a Web Doctor Viewer.
Authentication	Email, Google, Apple, and Phone Number + OTP.
Accounts	User account plus multiple family health profiles.
Documents	PDF, images, prescriptions, lab reports, radiology reports, and DICOM support.
Core Value	Medical timeline, medication tracking, symptom logging, AI doctor report, QR doctor access.
AI Strategy	Portable AI architecture: external APIs first, open-source/self-hosted option later.
Business Model	Hybrid model: subscription + credit consumption for AI-heavy features.
Excluded Feature	Water Intake Tracker is removed from this version.
Doctor Experience	Doctor web summary via QR or secure temporary access, read-only.
Vendor Request	Provide cost, timeline, team structure, stack recommendation, server sizing, maintenance plan, and full source code transfer.

2. Product Vision

To build the simplest personal health operating system that allows patients and families to keep a lifelong structured medical history and generate consultation-ready summaries for healthcare providers in seconds.

3. Business Context and Opportunity

Medical information is usually fragmented across labs, clinics, hospitals, messaging apps, email attachments, and paper documents. Patients often forget symptoms, treatment changes, or the timing of key events when they visit a doctor. This platform addresses that fragmentation by centralizing records and transforming them into a usable timeline and doctor-facing summary.

The product is especially valuable for chronic disease patients, multi-provider treatment journeys, busy adults, elderly users, and parents managing children's records.

4. Business Model

4.1 Model Type

The product uses a hybrid monetization model: subscription access plus credit-based usage for resource-intensive operations.

4.2 Why Hybrid

- Provides predictable recurring revenue.
- Keeps the entry price low for end users.
- Controls costs for AI processing, OCR, speech transcription, and doctor report generation.
- Prevents abuse or unsustainably heavy usage.

4.3 Proposed Plans

Plan	Price	Storage	Monthly Credits	Notes
Basic	\$1 / month	100 MB	1000	Entry package for early market adoption
Premium (future)	\$3 / month	1 GB	5000	Higher storage and heavier AI use
Top-up Credits	TBD	N/A	500 / 1000 / 5000	Additional usage packs as needed

4.4 Credit Consumption Examples

Feature	Illustrative Credit Cost	Comment
Upload medical report	10	File handling and indexing
AI document analysis / OCR	25	Structured extraction
Voice symptom transcription	5	Speech-to-text conversion
Lab comparison chart	15	Trend preparation and rendering
AI doctor report	50	Summary generation for selected period
Radiology summary	40	Applies only where summary is requested

5. Product Scope

This version covers patient and family health record management, medical timeline creation, medication and appointment tracking, symptom logging, doctor report generation, radiology and DICOM support, QR-based doctor access, notifications, and portable AI architecture planning.

This version explicitly excludes Water Intake Tracker. It also excludes medical diagnosis, automated treatment recommendations, and physician decision support.

6. Target Users

6.1 Primary Users

- Adults managing their own health records.
- Patients with chronic conditions and repeated follow-up testing.
- Busy users who need reminders and structured health organization.
- Elderly users with memory difficulties.
- Parents managing children's medical records.

6.2 Secondary Users

- Doctors reviewing structured medical summaries during consultations.
- Family caregivers supporting elderly or dependent patients.

7. Platforms

- Mobile Application for iOS.
- Mobile Application for Android.
- Web Doctor Viewer for browser-based review of generated patient summaries.

8. Authentication and Identity

- Email + Password login.
- Google Login.
- Apple Login.
- Phone Number + OTP.
- Session management and secure token handling.
- Ability to add user identity verification at onboarding or subscription stage if required by business or compliance policy.

9. Family Health Profiles

One account can manage multiple health profiles. The account holder must be able to switch among family members and access independent data for each profile.

- Each family profile has its own timeline, reports, medications, symptoms, measurements, and appointments.
- Permissions must ensure profile data is logically isolated within the same account.
- Doctor report generation must target a specific family profile and date range.

10. User Experience Model

The primary interaction model is conversational. The user should be able to interact with the system using text, voice, and simple quick-action buttons. This design is intended to keep the app simple for all age groups and especially accessible to users who are uncomfortable with complex multi-tab health applications.

10.1 Quick Actions

- Upload Report
- Add Medication
- Add Appointment
- Record Symptom
- Add Weight
- Generate Doctor Report

11. Core Functional Modules

11.1 Health Vault

- Upload and securely store medical reports, prescriptions, discharge summaries, lab results, radiology reports, and medical letters.
- Support PDF, JPG, PNG, and DICOM files.
- Associate each file with the correct family profile.
- Store original files and metadata such as upload date, report type, and optionally extracted text.

11.2 Health Timeline

- Generate a chronological history of all medical events.
- Include documents, symptoms, medications, appointments, weight measurements, blood pressure records, and user notes.
- Allow filtering by date range, profile, and event type.
- Medication history must support categories: current long-term, current short-term, historic long-term, historic short-term, and PRN.

11.3 Medication Manager

- Add medications manually.
- Optionally upload medication package images or prescription images to assist identification.
- Store dose, frequency, timing, duration, start date, and end date where applicable.
- Send reminders and track adherence and missed doses.

11.4 Appointment Manager

- Create appointments for doctor visits, lab tests, radiology appointments, and procedures.
- Store provider, location, date, time, and notes.
- Trigger reminder notifications and optionally suggest generating a doctor report before the consultation.

11.5 Symptom Logging

- Allow text-based symptom recording.
- Allow voice-based symptom recording.
- Convert voice entries into text via speech-to-text and store both the text representation and audio reference if needed.
- Attach timestamps and optionally severity.

11.6 Weight Tracking

- Allow entry of weight records over time.
- Display historical weight charts.
- Support BMI using stored height and weight.
- Support child growth use cases by retaining height records as needed.

11.7 Lab Comparison

- Compare the same laboratory test across multiple dates.
- Show tabular comparison and trend charts.
- Allow filtering by time period.
- Highlight notable changes where consistent values are available.

11.8 Radiology and Imaging

- Store radiology reports, image files, and DICOM studies.

- List imaging studies in profile view and doctor report.
- Allow opening or previewing available files in a supported viewer flow.
- Include radiology summaries only as descriptive organization support, not diagnosis.

11.9 Prescription Management

- Store prescription images and PDFs.
- Link prescriptions to timeline events and doctor reports.
- Optionally extract medication names from prescriptions where technically possible.

11.10 AI Doctor Summary Report

- Allow user to select a date range and generate a doctor-facing report.
- The report must summarize symptoms, medication changes, uploaded reports, lab trends, prescriptions, imaging, and measurements for the selected period.
- The output must be available in mobile view, web doctor view, and downloadable PDF.
- The report must remain tied to source records so users and doctors can inspect original documents if needed.

11.11 Doctor QR Access

- Allow patient to generate a secure QR code after preparing the doctor report.
- Doctor scans QR and opens a temporary read-only web view.
- Access should be time-limited and revocable where possible.
- The doctor should not need to install the application.

11.12 Notification System

- Medication reminders at configured times.
- Missed-dose reminders if user has not marked medication as taken.
- Appointment reminders, for example 24 hours, 3 hours, and 30 minutes before the appointment.
- Doctor visit preparation reminder prompting report generation.

11.13 Emergency Detection and Contact

- If the user records critical phrases such as severe chest pain or states they have fallen and cannot get up, the app should display an urgent prompt advising immediate medical attention.
- Emergency contacts may be stored so the user can quickly initiate outreach.
- The system must not present itself as an emergency service replacement.

12. AI Strategy and Portability

The product must be built with a portable AI architecture. The goal is to support rapid launch using third-party AI APIs and later reduce long-term cost by moving heavy workloads to self-hosted open-source models when user volume justifies the operational complexity.

12.1 Phase 1 – External AI APIs

- Use external APIs such as OpenAI, Azure OpenAI, Anthropic Claude, or equivalent providers.
- Primary use cases: document analysis, OCR support, speech-to-text, report summarization, symptom classification.

12.2 Phase 2 – Hybrid AI

- Allow some requests to continue through external APIs while selected heavy or repetitive tasks are moved to internal infrastructure.

12.3 Phase 3 – Open Source / Self-Hosted

- Support migration to models such as Llama, Mistral, Mixtral, or other suitable open-source options.
- All AI calls must be routed through an internal AI service layer so the provider can be switched without rewriting the mobile applications.

13. High-Level Technical Architecture

Suggested architecture for quotation purposes:

Layer	Purpose
Mobile Apps	Primary patient interface on iOS and Android
Web Doctor Viewer	Read-only doctor-facing access via QR or secure link
Backend API Layer	Business logic, profile management, records, reminders, sharing rules
AI Service Layer	Abstraction for external APIs and future self-hosted AI models
Structured Database	User, family, reports, medications, symptoms, measurements, appointments, AI reports
Secure File Storage	PDF, images, and DICOM storage
Notification Service	Push reminders and event notifications
Monitoring and Logging	Operational visibility and auditability

14. Data Model Overview

The final data model will be prepared by the vendor in implementation detail, but the quotation should assume the following domains:

- Users
- FamilyProfiles
- MedicalReports
- ReportAttachments
- Medications
- MedicationSchedules
- Symptoms
- Appointments
- Measurements
- AIReports
- DoctorAccessTokens / QR sessions
- Notifications
- AuditLogs
- BillingPlans
- CreditBalances
- CreditTransactions

15. Non-Functional Requirements

15.1 Security

- Encrypt data in transit and at rest.

- Use secure authentication and role-based access.
- Ensure family profile access is controlled correctly.
- Protect sensitive medical files in storage and delivery.
- Implement environment separation for development, staging, and production.

15.2 Performance and Scalability

- Target initial support for at least 10,000 users.
- Design upload handling for large medical files including DICOM.
- Ensure AI report generation and timeline loading perform within acceptable user-facing latency.

15.3 Availability

- Use cloud infrastructure with monitoring, backups, and incident visibility.

16. User Journeys

16.1 Primary Patient Journey

- User registers or logs in.
- Creates own profile and optionally family member profiles.
- Uploads medical reports and prescriptions.
- Adds medications and appointments.
- Logs symptoms and measurements over time.
- Before visiting a doctor, selects a date range and generates the doctor report.
- Shows QR to the doctor to open the web viewer.

16.2 Doctor Journey

- Doctor scans QR code during consultation.
- Opens read-only web summary.
- Reviews timeline, symptoms, medication history, lab comparisons, reports, prescriptions, and attached radiology files where available.

17. UI Screen Inventory

- Welcome / onboarding screens
- Login / signup
- Authentication method selection
- OTP verification
- Profile setup
- Family profile manager
- Home chat interface
- Quick action panel
- Timeline screen
- Report upload screen
- Report detail screen
- Medication list
- Add / edit medication
- Medication reminder confirmation
- Appointment list
- Add / edit appointment

- Symptom logging screen
- Voice symptom recording screen
- Weight entry screen
- Weight chart screen
- Lab comparison screen
- Radiology / imaging list
- Prescription list
- Generate doctor report screen
- Doctor report preview screen
- Doctor QR sharing screen
- Notifications center
- Settings and account screen

18. Feature Breakdown for Estimation

Module	Summary	Complexity	Illustrative Effort
Authentication	All login methods and secure sessions	Medium	2–3 weeks
Family Profiles	Multiple profiles under one account	Medium	2 weeks
Health Vault	Upload, classify, store, and retrieve documents	High	3–4 weeks
Timeline	Chronological event aggregation and filtering	High	3 weeks
Medication Manager	Schedules, reminders, adherence	Medium	2–3 weeks
Appointment Manager	Appointment creation and reminders	Low/Medium	1–2 weeks
Symptom Voice Logging	Voice capture and speech-to-text	High	2–3 weeks
Weight & BMI	Measurement entry and charts	Low	1 week
Lab Comparison	Repeated test comparisons and charts	High	2–3 weeks
Radiology & DICOM	Imaging support and file handling	High	3–4 weeks
Prescription Management	Prescription storage and indexing	Low/Medium	1 week
AI Doctor Report	Date-range summary generation	High	3–4 weeks
Doctor QR Access	Temporary doctor web session	Medium	1–2 weeks
Notifications	Medication and appointment reminders	Medium	2 weeks
Billing & Credits	Plans, balances, consumption, top-ups	High	2–3 weeks
Web Doctor Viewer	Read-only structured doctor summary	Medium/High	2–3 weeks

19. Infrastructure and Server Requirements for Phase 1

The vendor must provide a precise infrastructure recommendation and cost estimate. The following is the minimum expectation for quotation purposes and must be validated or improved by the vendor.

Component	Suggested Baseline	Purpose	Estimated Monthly Range
App/API Compute	2 application instances or equivalent container service	Run backend and business services	\$120–\$250
Managed Database	PostgreSQL managed service	Store structured product data	\$60–\$120
Object Storage	Secure file bucket for PDFs, images, DICOM	Medical file storage	\$50–\$180 depending on usage
CDN / File Delivery	Cloud CDN	Serve assets efficiently	\$20–\$50
AI API Budget	External AI provider usage	OCR / STT / summaries	\$100–\$400 depending on usage
Monitoring / Logs	Cloud monitoring stack	Operational visibility	\$20–\$60
Backups	Automated snapshots and retention	Recovery and continuity	Included or \$20–\$50

Expected early-stage infrastructure cost range: approximately \$300 to \$1,000 per month depending on storage volume, AI usage, and file traffic.

The vendor should specify whether GPU infrastructure is not required in Phase 1 because AI uses external APIs, and should separately state what would be needed for a future self-hosted model path.

19.1 Vendor Must Recommend

- Preferred cloud provider (AWS / GCP / Azure / other) with justification.
- Recommended server sizing for phase 1.
- Expected storage strategy for DICOM and large reports.
- Backup, monitoring, and security controls.
- Migration considerations for later open-source AI hosting.

20. Development Phases

20.1 Phase 1 – MVP

- Authentication
- Family profiles
- Health vault
- Timeline basic
- Medication manager
- Appointment manager
- Weight tracking
- Basic notifications
- Web doctor viewer foundation

20.2 Phase 2 – Smart Health Layer

- Voice symptom logging
- Prescription organization
- Lab comparison
- Radiology and DICOM support
- Doctor QR access

20.3 Phase 3 – AI Expansion

- Full AI doctor report generation

- Portable AI abstraction layer refinement
- Hybrid / self-hosted AI migration readiness
- Advanced extraction and summarization improvements

21. Vendor Deliverables

- Mobile applications for iOS and Android.
- Web Doctor Viewer.
- Backend APIs and business logic.
- Database schema and migration scripts.
- AI integration layer.
- Cloud deployment instructions.
- Environment setup details.
- Technical documentation.
- Handover package.

22. Vendor Quotation Request

Vendors must provide a quotation that includes all of the following:

- Detailed cost estimate by module or phase.
- Development timeline and milestone plan.
- Team structure and roles.
- Recommended technology stack.
- Server / cloud recommendation for phase 1.
- Post-launch support and maintenance proposal.
- Assumptions, exclusions, and dependencies.

23. Intellectual Property and Code Ownership

All source code, documentation, infrastructure code, design files, API definitions, and project-specific implementation outputs must become the property of the client upon payment according to contract terms.

- The client must receive full source code and deployment instructions.
- The client must have the right to host, modify, extend, or transfer the project.
- The vendor must disclose any third-party licensed components and their restrictions.
- The vendor must not lock the client into vendor-only hosting or inaccessible proprietary code without explicit written approval.

24. Out of Scope

- Medical diagnosis
- Treatment recommendation engine
- Prescribing medication
- Replacing healthcare professionals
- Water Intake Tracker

25. Final Notes for Estimation

This document is intentionally detailed so that vendors can provide a realistic quotation. If any assumptions are needed, the vendor must clearly list them in the commercial proposal. Any recommendations for improving security, scalability, AI cost control, or implementation sequencing are welcome, provided they preserve the core product intent described in this document.